

SECTION 2 - GENERAL TECHNICAL REQUIREMENTS

List of Subsections

	Page No.
SUBSECTION - 2.1 BID DRAWINGS	7
SUBSECTION - 2.2 DATA AND DRAWING SUBMISSION	8
SUBSECTION - 2.3 SCHEDULES	19
SUBSECTION - 2.4 PROGRESS REPORTS	22
SUBSECTION - 2.5 CODES AND STANDARDS	25
SUBSECTION - 2.6 PERFORMANCE TESTING, GUARANTEES AND FIRST INSPECTION	28
SUBSECTION - 2.7 QUALITY ASSURANCE / QUALITY CONTROL (QA/QC)	36
SUBSECTION - 2.8 PLANT IDENTIFICATION	43
SUBSECTION - 2.9 GENERAL NOISE CRITERIA	46
SUBSECTION - 2.10 INSTRUCTION MANUALS	49
SUBSECTION - 2.11 SPARE PARTS, TOOLS, TEST EQUIPMENT AND CONSUMABLES	53
SUBSECTION - 2.12 PACKAGING INSTRUCTIONS AND STORAGE	59
SUBSECTION - 2.13 TRAINING	66
SUBSECTION - 2.14 SEISMIC REQUIREMENTS	69
SUBSECTION - 2.15 UNITS OF MEASUREMENTS AND SYMBOLS	71
SUBSECTION - 2.16 ENVIROMENTAL CONSIDERATIONS	74
SUBSECTION - 2.17 FLUE GAS ANALYSIS	84

SECTION 2 GENERAL TECHNICAL REQUIREMENTS

TABLE OF CONTENTS

	Page No.
SUBSECTION - 2.1 BID DRAWINGS	7
2.1.1 GENERAL	7
2.1.2 DRAWINGS	7
SUBSECTION - 2.2 DATA AND DRAWING SUBMISSION	8
2.2.1 GENERAL	8
2.2.2 DATA AND DRAWINGS REQUIRED WITH PROPOSAL	8
2.2.2.1 General Arrangements	8
2.2.2.2 Description of All Plant Equipment and Systems	9
2.2.2.3 Diagrams	9
2.2.2.4 Technical Data Sheets	9
2.2.2.5 Performance Data and Curves	9
2.2.2.6 Quality Assurance and Quality Control Data	10
2.2.2.7 Proposed Construction Schedule	10
2.2.2.8 Statements of Design Intent for Structure, Civil and Building Works	10
2.2.3 DATA AND DRAWINGS TO BE ISSUED BY THE BIDDER .	10
2.2.3.1 General	10
2.2.3.2 Submission and Approval	11
2.2.3.3 General Data and Drawings:	13
2.2.3.4 Data and Drawings to be submitted within Two (2) Month	13
2.2.3.5 Data and Drawings to be submitted within Three (3) Months	14
2.2.3.6 Data and Drawings to be submitted within four (4) Months	15
2.2.3.7 Data and Drawings to be submitted within Five (5) Months.....	16
2.2.3.8 Revision of Bidder's Drawings at Site	16
2.2.3.9 As Built Drawings	16

2.2.3.10 Contract Drawings.....	16
2.2.3.11 Erection Drawings	17
2.2.3.12 Drawings and Documents Management System	17
2.2.3.13 Maintenance Management System.....	18
SUBSECTION - 2.3 SCHEDULES	19
2.3.1. PROJECT SCHEDULE	19
2.3.2. CONTRACT SCHEDULE	20
SUBSECTION - 2.4 PROGRESS REPORTS	22
2.4.1. SCHEDULE WORK.....	22
2.4.2. COST CONTROL	23
SUBSECTION - 2.5 CODES AND STANDARDS	25
2.5.1 GENERAL	25
2.5.2 LIST OF REFERENCES	25
SUBSECTION - 2.6 PERFORMANCE TESTING, GUARANTEES AND FIRST INSPECTION	28
2.6.1 GENERAL	28
2.6.2 SCOPE	28
2.6.3 TEST PROCEDURES	30
2.6.4 EQUIPMENT PERFORMANCE	30
2.6.5 PLANT GUARANTEES	31
2.6.5.1 Guarantee Conditions:.....	31
2.6.5.2 Gross Power Generation	32
2.6.5.3 Auxiliary Power Consumption	32
2.6.5.4 Correction Curve	32
2.6.5.5 Bonafide Tests	32
2.6.5.6 Test Period and Test Condition	33
2.6.5.7 Instrumentation accuracy	33
2.6.6 TEST BEFORE SHIPMENT	34
2.6.7 TEST RECORDS	34

2.6.8	GUARANTEE AND REJECTION.....	34
2.6.9	FIRST INSPECTION.....	35
SUBSECTION - 2.7 QUALITY ASSURANCE / QUALITY CONTROL (QA/QC).....		36
2.7.1.	STANDARD QUALITY ASSURANCE/QUALITY CONTROL	36
2.7.1.1	Bidder's Quality Assurance/Quality Control System Description	36
2.7.1.2	Detailed Quality Control Program.....	36
2.7.1.3	Overall Inspection Test Program.....	37
2.7.1.4	Sub Bidder's QA/OC System and Procedures	37
2.7.1.5	Documentation Required	37
2.7.1.6	Contract Non-conformances	38
2.7.1.7	Radiographs	38
2.7.1.8	Legibility of Documentation.....	38
2.7.1.9	Honoring of Invoices.....	38
2.7.1.10	Completion Letter	38
2.7.2.	STANDARD PROCEDURE FOR PACKAGING AND SHIPMENT OF RADIOGRAPHS	39
2.7.2.1	General	39
2.7.2.2	Packaging of Radiographs	39
2.7.2.3	Shipments of Radiographs	39
2.7.3.	SHOP INSPECTIONS	40
2.7.3.1	Shop Inspection and Test Program / Inspection and Test Time Schedule	40
2.7.3.2	Shop Inspection Schedule	40
2.7.3.3	Shop Inspection Procedures.....	41
2.7.3.4	Shop Inspection Notification	41
2.7.3.5	Document Approval Status	41
SUBSECTION - 2.8 PLANT IDENTIFICATION		43
2.8.1.	GENERAL	43

2.8.2.	NUMBERING SYSTEM.....	43
2.8.3.	NAMEPLATES AND LABELS	44
SUBSECTION - 2.9 GENERAL NOISE CRITERIA		46
2.9.1	GENERAL	46
2.9.2	CODES AND STANDARDS.....	46
2.9.3	GENERAL DETAILED REQUIREMENTS	46
SUBSECTION - 2.10 INSTRUCTION MANUALS.....		49
2.10.1.	GENERAL	49
2.10.2.	DESIGN MANUALS	50
2.10.3.	OPERATIONS MANUALS.....	50
2.10.4.	MAINTENANCE MANUALS	51
2.10.5.	DIGITAL SYSTEMS MANUAL	52
SUBSECTION - 2.11 SPARE PARTS, TOOLS, TEST EQUIPMENT AND CONSUMABLES		53
2.11.1.	SPARE PARTS	53
2.11.2.	TOOLS AND TEST EQUIPMENT	55
2.11.3.	CONSUMABLE PARTS AND MATERIALS.....	57
SUBSECTION - 2.12 PACKAGING INSTRUCTIONS AND STORAGE		59
2.12.1	GENERAL	59
2.12.2	PACKAGING INSTRUCTIONS.....	59
2.12.3	RECEIVING	62
2.12.4	STORAGE	63
2.12.5	SPECIAL PROTECTIVE MEASURES	64
2.12.6	HEATER	64
2.12.7	CABLE ON REELS.....	64
2.12.8	INSTALLED EQUIPMENT AND MATERIAL	65
SUBSECTION - 2.13 TRAINING		66
2.13.1	GENERAL	66
2.13.2	DEFINITIONS	66

2.13.2.1 Overview	66
2.13.2.2 Maintenance	67
2.13.2.3 Troubleshoot	67
2.13.3 DETAILED REQUIREMENTS	67
SUBSECTION - 2.14 SEISMIC REQUIREMENTS	69
2.14.1 SCOPE	69
2.14.2 CODES AND STANDARDS	69
2.14.3 BUILDINGS AND STRUCTURES HIGHER THAN 40 m	69
SUBSECTION - 2.15 UNITS OF MEASUREMENTS AND SYMBOLS	71
2.15.1 UNITS OF MEASUREMENTS AND SYMBOLS	71
SUBSECTION - 2.16 ENVIROMENTAL CONSIDERATIONS	74
2.16.1 GENERAL	74
2.16.2 ENVIRONMENTAL LEGAL AND REGULATORY	74
2.16.3 ENVIRONMENTAL REQUIRMENT	75
SUBSECTION - 2.17 FUEL ANALYSIS	82
2.17.1 GENERAL	82

SUBSECTION - 2.1 BID DRAWINGS

2.1.1 GENERAL

Owner's Drawings included in **Part 6**, are Bid drawings.

2.1.2 DRAWINGS

Drawing issued with this Specification known as Bid drawings show the arrangements of Plant and Systems required by the Owner.

The Bidder is requested to bid the Works in accordance with the Owner's arrangements.

If the arrangements of plant proposed by the Bidder are different from the Owner's arrangements, such differences shall be fully explained and justified in the proposal.

The Bidder shall arrange the equipment, panels, piping and other system components in such to give and ample maintenance space.

The overall flow diagrams shown in the Bid Drawings are preliminary only. The Bidder shall make all complete P&IDs to meet the requirement of standards and codes applicable to associated system.

(Last page of Subsection 2.1)

SUBSECTION - 2.2 DATA AND DRAWING SUBMISSION

2.2.1 GENERAL

The Bidder shall submit with his proposal all data, information and drawings specified herein. Data and curves related to the performance of the plant items and equipment shall be a part of the Contract.

Following the award of the Contract, the Bidder shall prepare and provide to the Owner revised and certified Bid Proposal documents including all data and drawings mentioned above being correct in every detail, as well as all documents, data, calculations and drawings showing the detailed design performed by the Bidder, to be approved by the Owner.

2.2.2 DATA AND DRAWINGS REQUIRED WITH PROPOSAL

The Bidder shall submit at least the following data and drawings with his Proposal.

2.2.2.1 General Arrangements

Layout with elevations and sections showing the plant arrangement, principal dimension, parts to be assembled on Site, and shall also clearly indicate the laydown areas and major component removal routes. The general plant arrangement drawing shall include sufficient plan and elevation views to show the arrangement of all equipment both horizontally and vertically in relation to each other.

- General arrangement of all significant items of plant to be supplied, including cross section drawings and details.
- General arrangement of all control panels, switchboards, motors and power control centers, and transformers. The layout of instruments and control functions shall be shown.
- Proposed foundation concept for all major plant equipment and buildings structures supplied under this Contract.
- Layout and conceptual arrangement showing routes of principal power cable, control cable, main interconnecting piping and ducts.

2.2.2.2 Description of All Plant Equipment and Systems

- Description of all plant equipment and systems including start-up and shut-down procedures, operation mode, control philosophy, and special features of the equipment and systems.

2.2.2.3 Diagrams

- Process and flow diagram for all systems supplied under this Contract.
- Functional logic diagrams representing the major components and system of the plant. The diagrams shall be in the form of standardized P & IDs and shall be completed in every detail, represent of equipment and piping system including valves, instruments, vents, traps, drains, sample connections, etc.
- Electrical single line diagrams, of Substation and Plant electrical system, including protection diagrams.
- Heat Balance Diagram for conditions of:
 - MCR load conditions for each combined cycle block 2, 3 & 4
 - Part load with 2 Gas Turbines at base load of the combined cycle plant for each combined cycle block 3 & 4
 - Part load with 1 Gas Turbine at base load for each combined cycle blocks

2.2.2.4 Technical Data Sheets

- Completed data sheets as included in Schedule 7

2.2.2.5 Performance Data and Curves

Performance Data and Curves for the Add On Blocks supplied under this Contract which shall include, but not be limited to, the followings:

- Steam turbine performance curves
- Generator and excitation performance curves
- Correction curves due to ambient and seawater temperature, pressure, level, frequency changes, etc.
- Start-up, loading and normal shut-down curves

2.2.2.6 Quality Assurance and Quality Control Data

- Quality assurance and quality control data

2.2.2.7 Proposed Construction Schedule

- Proposed construction schedule

2.2.2.8 Statements of Design Intent for Structure, Civil and Building Works

- Statements of design intent for civil works which define the extent and purpose of works to be designed and include description of the philosophies that governs the design.

2.2.3 DATA AND DRAWINGS TO BE ISSUED BY THE BIDDER

2.2.3.1 General

Following the award of the Contract, the Bidder shall submit all drawings, schedules, calculations, documents, data and information necessary for the proper supervision by the Owner of the work involved.

Such data and drawings submittals shall include those specified and shall be sufficiently comprehensive to fully establish that all parts and procedures to be used in performing the Contract Works comply with the objective of the Contract and with requirements for installation, operation and maintenance.

The Owner shall have the right to require the Bidder to submit such additional information as may reasonably be required.

All drawings and documents prepared by the Bidder or his sub-Bidders for this Contract shall bear an approved contract reference and title block, and Owner's reference drawing number.

All documentation whether letters or drawings, shall show the project title and number and shall be on ISO standard A-size sheets with the normal maximum size of drawing is A0 size.

All documentation, whether letters, lists or drawings, shall be made available in electronic form.

All drawings shall be to scale and fully detailed and dimensioned. The material of which each part is to be constructed shall be indicated. Each drawing shall be

clearly stamped to indicate the applicable status e.g. For Bid, Preliminary, For Information, For Approval, For Construction, and As Built.

Electrical schematic and wiring diagrams shall be arranged such that terminal blocks and terminal information is clearly shown and indicating incoming and outgoing cables and the location of the remote end of such cables. Circuit diagrams of relevant electrical equipment shall be provided sufficient for comprehensive fault finding.

2.2.3.2 Submission and Approval

Before the commencement of construction/fabrication of each items of the plant, the Bidder shall submit for approval by the Owner the design drawings with associated documents, design criteria, calculations.

To catch up drawing approval progress, the Owner will carry out drawing review approval at the Bidder design home office for duration a long two weeks.

Drawing review approval will be done several times in overseas. The Bidder shall propose the list of major drawings to be reviewed at overseas and the cost of all expenses shall be included in the bid proposal price.

Documents submitted for approval shall provide sufficient information to the Owner to enable the Owner checking that all items and equipment intended to be put into construction/fabrication by the Bidder comply with the Contract.

Detailed fabrication drawings will not be normally subject to Owner's approval, however the Bidder shall make them available to the Owner if so requested.

The Bidder shall submit six (6) copies of drawing to Owners for approval.

If the Owner is satisfied with the drawing, one (1) copy will be returned to the Bidder marked with "Approved" stamp within four (4) weeks for "normal" documents and two (2) weeks for "urgent" documents. If the Owner is not satisfied with the drawing, one (1) copy stamped "Approved as Noted" or "Not Approved" will be returned to the Bidder within said period, with comments marked thereon and the drawing shall be amended, the revision letter affixed to the Drawing Number revised and re-submitted for approval.

Revised drawing shall have indicated on them the nature of the change, the reason for the revision and the location of the revision.

When submitting drawing for approval, including any prepared by a sub-Bidder, the Bidder shall certify that he has fully examined such drawings and that they comply with the requirements of the Contract.

Approval of a drawing by the Owner shall in no way relieve the Bidder of his responsibilities under the Contract.

The Owner's comments on these drawings and data shall be resolved by the Bidder and returned to the Owner within two (2) weeks from the date of the Owner's comments.

Upon receipt of approval by the Owner, the Bidder shall arrange for the "revision column" of the drawings to be marked "approved by Owner on (date)", revised the issue letter of number, and forward four (4) copies to the Owner for his use.

If the Bidder intends to divide or subcontract any part of the work he shall ensure that the design is fully coordinated between all the parties involved prior to submitted of relevant drawings and documents for Owner's review and approval. For this purpose he shall designate an individual or group of individuals charged with the execution of this coordination work. The Bidder shall provide an organization chart clearly indicating the interfaces involved and identifying the individual to whom the Owner can refer as the "single point of contact" for this design coordination effort. The Bidder shall have their engineering representative in their Jakarta office, available to the Owner for technical discussions, during the entire duration of the Project. The subject engineering representatives shall cover mechanical, electrical, civil/structural and instrumentation and control engineering disciplines, and shall have full knowledge of the design requirements with full authority to resolve any discrepancies, and to act to implement all design resolutions and Owner's directives. The time required to consult to Bidder's head office shall be minimized.

No works shall be made without Owner's prior approval to the drawings, calculations or procedures selected by the Owner.

Any work done before the review of the drawings and data shall be at the Bidder's risk and any necessary design changes to comply with the requirements and objectives of the Contract shall be made at no additional cost to the Owner or without any delay to the Project execution.

The detailed list comprising all drawings and documents the Bidder will produce with the information which drawings or documents will have to be submitted for approval, information or any other purpose shall among others contain following drawings, data and comments:

2.2.3.3 General Data and Drawings:

- A design criteria of structural and architectural, mechanical and electrical equipment selection and control work.
- Description of cleaning, checking and testing procedure.
- For piping and valves, materials fully identified by referencing the appropriate Standard of material, specification number and grade or each size.
- Specifications covering construction and materials used in the equipment. Also description of special equipment, if included, for dismantling and assembly of equipment.
- An itemized list of all tools that he will furnish with the equipment that are required for maintenance and repair of equipment.
- Calculations for all flow elements. Information shall include data for both design and operating conditions. Pressure drop of the element shall also be included.
- Bidder shall submit general erection procedures and startup and commissioning procedures for all major equipment as requested.
- Any other information required by the Owner.

The Owner will not necessarily examine in detail all submittals forwarded to him by the Bidder and may at his option require submittals to be subject to his review or regard them as for information and record purposes.

The Bidder shall be responsible for any discrepancies, errors, or omissions on the drawings and other particulars supplied by him whether or not such drawings and particulars have been reviewed by the Owner.

The Owner shall require the Bidder to make any changes to the drawing and data which may be necessary to make the Works conform to the intent of the Contract until full acceptance of the equipment by the Owner.

2.2.3.4 Data and Drawings to be submitted within Two (2) Month

Within two (2) month of the date of the Contract Award, the Bidder shall submit the following:

- Heat Balance Diagram for conditions of:
 - MCR load conditions for each combined cycle block 2, 3 &4
 - Part load with 2 Gas Turbines at base load of the combined cycle plant for each combined cycle block 3&4
 - Part load with 1 Gas Turbine at base load for both fuels for each combined cycle blocks
- General description and technical concept of all major equipment item.
- Contract network schedule.
- Drawing list summarizing all existing and proposed drawings of principal items intended for submission to the Owner by the Bidder, together with the submission dates.
- Overall layout and general arrangement of all major plant giving position and dimension of main buildings and components.
- Single line diagrams of main electrical connections.
- Contract program
- Inspection Test Plan (ITP) and quality. Test procedures at factory for all equipment supplied under this Contract within (2) weeks prior to (FAT).
- Equipment and system identification numbers using standardized KKS (Kraftwerks Kennzeichen System/Identification system for Power Plant) Number System.

2.2.3.5 Data and Drawings to be submitted within Three (3) Months

Within three (3) months of the date of Contract Award, the Bidder shall submit the following:

- Detailed foundation/civil drawings providing all necessary information for the detailed civil works and building design for all plant, together with a schedule of built-in items.
- Complete equipment list of all plant items and equipment to be supplied together with the name of the manufacturer or sub-Bidder, country or origin, and FOB delivery date.

- Access and supporting steel arrangement drawings
- Piping layout drawings
- Cables ,conduits and tray routing
- Plant cable diagrams and diagrams of connections for all plant items, marshalling boxes and control panels, and itemized cable schedule
- Electrical schematic diagrams, including power, control, indication, protection and alarms.
- Instrument lists, functional logic diagrams for control, alarm and sequence interlocks.
- Schedule of all AC and DC loads.
- Drawing list summarizing all existing and proposed drawings of principal items intended for submission to the Owner by the Bidder together with submission date.

2.2.3.6 Data and Drawings to be submitted within four (4) Months

Within four (4) months of the date of the Contract Award, the Bidder shall submit the following:

- Piping layout drawings
- Cables and Conduits routing
- Plant cable diagram and diagram of connections for all plant items, marshalling boxes and control panel and itemize cable schedule.
- Electrical schematic diagram including power, control, indication, protection and alarms.
- Detailed general arrangement drawings for all major plant items.
- All welding procedures (both at Site (4 Weeks prior erection start) and during manufacture)
- Access and supporting steel arrangement drawing
- Schedules of consumable items, lubricants and chemicals
- Standard format schedules for orifice plates, all valves, control valves, safety valves, strainers, motors and drivers, etc. using the approved KKS number.
- Detailed drawings of steam turbine buildings including services

2.2.3.7 Data and Drawings to be submitted within Five (5) Months

Within five (5) months of the date of the Contract Award, the Bidder shall submit the following:

- Full manufacturer's data sheet on every type of device, equipment, or module which the Bidder proposed to utilize in the instrumentation and control system including computer hardware and software.
- Functional group control diagrams
- Schedules of instruments, valves, input and output signals (analogue and binary), using the approved system of plant item referencing and nomenclature
- Evidence of submission of plans for approval, where necessary, to local authorities
- Detailed dynamic and steady state response data.

2.2.3.8 Revision of Bidder's Drawings at Site

During the erection and commissioning period it may be necessary on the approval of the Owner to make revisions to already approved construction drawings. In such case the Bidder shall provide to the Owner at Site without delay two (2) marked-up prints detailing any revision made. Normal revisions of these drawings shall then be made and issued to the Owner as described earlier.

2.2.3.9 As Built Drawings

Following the installation of the equipment, the Bidder shall revise his drawings as necessary to show the "as built" status. The Bidder shall send three (3) copies of the "as built" drawings to the Owner for approval.

All As Built drawings shall be made available on DVD as pdf files.

2.2.3.10 Contract Drawings

After approval of the "as-built" drawings, the Bidder shall submit, for final record purposes, sets of all approved drawings as follows:

- One (1) set of electronics copy in the DVDs storage in PDF copy format
- One (1) set of full size paper print

- Two (2) sets of A3 size paper prints bound into albums

These sets of drawings shall be submitted as a single submission to the Owner at PLN.

2.2.3.11 Erection Drawings

It is the Bidder's responsibility to ensure that field erection personnel have in their possession the latest available erection drawings of the type and quantity needed to perform the Works.

2.2.3.12 Drawings and Documents Management System

The Bidder shall provide a Drawing and Document Management System which includes the application of walk-through model of the Plant. All for the project required drawings shall be generated using a system software as for example "Intergraph" PDS system or equivalent (PDMS).

The system shall include integrated 3D modules which generates the required documents for engineering, fabrication/construction and installation/ erection tasks such as:

- Process flow diagrams
- Piping and Instrumentation Diagram
- Instrumentation Engineering Document
- Piping documentation (isometrics etc.)
- Piping support modeler and explorer
- Interference checking
- Part lists
- Engineering data base, etc.

System Performance

The system must be performed for extraction of the piping data required for fabrication and installation such as: pipe size, wall thickness schedule, material type, pipe length, fitting size and type, valve type, bolting requirements component orientation, etc.

All individual spool pieces, whether it is to be fabricated in the shop or in the field, are specified in the system file, as such, and can be individually viewed by either the shop fabricator, the field fabricator or the installation Bidder for relevant data retrieval.

For the main pipeline systems the pipe supports and type will be shown in the system file.

2.2.3.13 Maintenance Management System

For the purpose of maintenance control and planning during operation of the Plant equipment, the Bidder shall provide an integrated maintenance management system including all required hardware and software.

The Maintenance Management System shall be suitable to plan and manage ongoing maintenance and repair operation and to track related labor, parts, and costs.

Furthermore, the system shall provide in separate modules features to plan, organize and control:

- work orders
- inspections
- preventive maintenance
- maintenance resources
- deployment of maintenance personnel
- etc.

Additionally, the Bidder shall ensure that the equipment vendors (suppliers to the Bidder) shall complete standardized equipment data forms and technical form.

(Last page of Subsection 2.2)

SUBSECTION - 2.3 SCHEDULES

2.3.1. PROJECT SCHEDULE

The Bidder shall provide with his Bid Proposal as required in **Part 5 Schedule 4**, a schedule in network form based on the dates given in **Part 5, Schedule 4** and the Project Outline Schedule. The network shall be based on Bid Proposal information and include:

- Dates for ordering of special and/or long delivery items.
- Dates for issue of drawings for planning purposes.
- Dates for start and completion of design engineering for all disciplines
- Dates for submittal to and return after final review by the Owner of all drawings and documents listed in Part 4, Subsection 2.2, of the specification.
- Dates for issue of all drawings for manufacture and construction purposes requiring the Owner's review.
- Dates for the placing of orders for material, receipt of material, fabrication and manufacture, factory testing and shipment.
- Dates for soil investigation.
- Dates showing start and completion of site construction of each section of the permanent works.
- Dates for periods when equipment is available for work by the Bidder
- Dates for submittal of design, operation and maintenance manuals for approval.
- Dates for submittal of commissioning check lists and detailed schedules for approval.
- Dates for submittal of complete schedules for all manufactured items

Activities shall be timed in week units except for commissioning or similar detailed program which have activities specified in days.

Activities for which it is intended to operate multiple shift working shall be clearly defined.

Each system as itemized in **Part 5, Schedule 1** shall be included.

The Bidder shall complete **Part 5, Schedule 3** with Schedule of site personnel and Constructional Plant to be employed on site.

2.3.2. CONTRACT SCHEDULE

The Schedule provided in accordance with Clause 1 above shall be revised in detail by the Bidder to reflect any changes in scope of Work or scheduled dates agreed during Contract negotiations and to a format acceptable to the Owner suitable for computer application.

The Bidder shall utilize a precedence method scheduling system which shall be compatible with the Owner's PRIMAVERA or Microsoft Project program. The details of the input format and the input required to the Bidder will be provided during the Contract negotiation.

The revised schedule shall be submitted to the Owner as part of the Contract Document to be signed and will become the Contract Schedule.

The Contract Schedule shall be revised by the Bidder to incorporate any variations in the scope of work and in the schedule agreed during the Contract. The revised schedule shall be submitted promptly to the Owner for review and approval.

The Bidder shall submit 6 (six) copies of all programs (original or revised issues) to the Owner for approval. After approval copies shall be supplied in accordance with the procedures specified in Part 4, Subsection 2.2 of the Specification.

Following approval of the Contract Schedule, detailed programs for all Work to be executed during the first twenty-four (24) weeks of the Contract shall be submitted.

These programs, which shall embrace design, procurement, manufacture, testing, shipping and erection on Site, shall be based on the Contract Schedule and shall be used as target programs possibly subject to revision.

The detailed programs shall be submitted to the Owner at twelve (12) week intervals, updated to reflect the changes that have accrued in the previous twelve

(12) weeks and to show the detailed program of Work for the next twenty four (24) weeks.

(Last page of Subsection 2.3)

SUBSECTION - 2.4 PROGRESS REPORTS

2.4.1. SCHEDULE WORK

The Bidder shall prepare and submit to the Owner within 15 days after the first day of every month, a written Monthly Progress Report summarizing the progress of the various sections of the works both in the factory and at Site. Such progress reports shall indicate accurately the status of different activities covering design, major procurement, manufacture, factory tests, shipping, erection, testing and commissioning and shall be related to key dates identified in the programs referred to in **Part 4, Subsection 2.3**. The report shall also include data on labor strength and equipment employed at the Site.

The Monthly Report shall include the Contract Schedule indicating progress and individual component progress made in the preceding month.

Such schedules shall show cumulative progress towards scheduled completion (expressed as a percentage) of all items shown in the Contract Schedule.

The report shall indicate the degree of criticality in each section of the work together with the slippage or impending slippage on any key event and shall be directly related to the Contract Schedule and supporting detailed network for sections of Work.

Six (6) copies of the Monthly Progress Report shall be submitted to the Owner in English in a format approved by the Owner.

The Bidder's Site Manager shall prepare and submit to the Owner six (6) copies of a weekly summary report covering all the Site activities. This report shall include projected Work activities for at least 14 Days after those being reported upon.

In addition this report shall include a weekly Site labor return giving expatriate and local labor and each subBidder's labor broken down into trades. Full details of Site labor disputes (or off-Site disputes affecting the Contract) shall be reported to the Owner immediately. The weekly statement shall give details of all Constructional Plant, machinery, offices and materials and include Site and weather conditions which may affect the progress of the Works. This weekly

summary report shall also include percent complete information by discipline and bulk commodities installation information: cubic meters of concrete, linear meters of cable, tons of structural steel, cable terminations, etc.

Report shall also include copy of the latest cash flows marked up to reflect Bidder's expenditures up to the reporting date.

The following daily and weekly reports on schedule shall be provided by the Bidder:

- The Bidder shall prepare and submit to the Owner on a daily basis and no later than 10:00 hours a daily work plan manpower report for each lower tier Bidder.
- A four (4) weeks look ahead schedule shall be prepared and submitted to the Owner at the start of each work week. This schedule shall show the detailed works completed in the previous two (2) weeks, detailed works to be completed this week and up to four (4) weeks in the future.

2.4.2. COST CONTROL

Bidder shall submit, with his Bid Proposal a time-phased, estimated cash flow on a month-by-month basis. The payments for labor and materials shall be separated and shown for each month, reflecting the estimated monthly costs for the Contract throughout its scheduled life.

The Bidder shall revise the cash flows submitted in the Bid proposal (see Clause above) to incorporate any variations in the Scope of Work and schedule dates agreed during Contract Negotiations and comply with the Owner's Work Breakdown Structure and code of accounts. The revised format and cash flows shall be submitted to the Owner for his review and comments immediately after Contract Award.

The cash flows agreed above shall be revised promptly by the Bidder to incorporate any relevant approved variations in the Contract Price and Contract Schedule as they occur.

The Monthly Progress Report required in Clause 1 above shall include a copy of the latest cash flows marked up to reflect Bidder's expenditures up to the reporting date.

Five (5) copies of the Monthly Cost Report shall be submitted to the Owner in English in a format approved by the Owner.

(Last page of Subsection 2.4)

SUBSECTION - 2.5 CODES AND STANDARDS

CDA
E-003

2.5.1 GENERAL

The codes, standards and specifications referenced in this Document (including addenda, amendments and errata) shall govern in all cases where references thereto are made. In case of conflict between these codes (or standards or specifications) and the requirements specified in this Document, the latter shall govern.

The codes, standards or specifications applied in each case shall be the latest revision adopted and published at the bid closing date. Any conflict between standards shall be referred to the Owner who will determine which standard shall govern and the Owner's decision will be final.

The Bidder shall state in his Bid Proposal the codes and standards which are not in conformance with this Bid Document together with a statement that in the Bidder's opinion the alternatives are equivalent or superior to those specified. If, during the evaluation period, the Owner has reasons to doubt the adequacy of alternative codes and standards, the Owner may request that information be submitted substantiating that Bidder's codes and standards are equivalent or superior to those specified.

After the Award of the Contract, the Owner may request that the Bidder submit selected alternative codes and standards for review and approval. Such codes and standards shall be submitted in the English language. No purchasing, fabrication or manufacture shall proceed until the codes and standards are approved in writing by the Owner.

2.5.2 LIST OF REFERENCES

In general the following codes and standards shall be used:

Reference Abbreviation	Name
AABC	Associated Air Balance Council
ACI	American Concrete Institute
ADC	Air Diffusion Council

Reference Abbreviation	Name
ABMA	The American Bearing Manufacturers Association
AGMA	American Gear Manufacturers Association
AISC	American Institute of Steel Construction, Inc.
AISE	Association of Iron and Steel Engineers
AISI	American Iron and Steel Institute
AMCA	Air Movement and Control Association International, Inc.
ANSI	American National Standards Institute
AIJ	Architectural Institute of Japan
API	American Petroleum Institute
ARI	Air Conditioning and Refrigeration Institute
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc.
ASME	American Society of Mechanical Engineers
ASNT	American Society for Nondestructive Testing
ASTM	American Society for Testing and Materials
AWS	American Welding Society
AWWA	American Water Works Association
BS	British Standard
CEIAC	Coastal Engineering Defense Information Analysis Center
CGA	Compressed Gas Association
CMAA	Crane Manufacturers Association of America
DIN	German Institute for Standardization
DPU	Departemen Pekerjaan Umum, Ditjen Cipta Karya Direktorat Penyelidikan Masalah Bangunan
EN	European Committee for Standardization
FED SPEC	Federal Specifications, Standards and Commercial Item Descriptions
FM	Factory Mutual Insurance Co.
HEI	Heat Exchange Institute
HIS	Hydraulic Institute Standards
HMI	Hoists Manufacturer's Institute
ICEA	Insulated Cable Engineers Association
IEC	International Electro-technical Commission
IEEE	Institute of Electrical and Electronics Engineers
ISA	Instrument Society of America
ISO	The International Organization for Standardization
JASS	Japanese Architectural Standard Specification

Reference Abbreviation	Name
JIS	Japan Industrial Standard
MBMA	Metal Building Manufacturers Association, USA
MSS	Manufacturers Standardization Society of The Valve and Fitting Industry, Inc.
NAAMM	National Association of Architectural Metal Manufacturers
NACE	National Association of Corrosion Engineers
NBS	National Bureau of Standards, USA
NEBB	National Environmental Balancing Bureau
NEMA	National Electrical Manufacturers Association
NERC	North American Electric Reliability Council
NFPA	National Fire Protection Association
OSHA	Occupational Safety and Health Act
PFI	Pipe Fabricators Institute
PUIL	Peraturan Umum Instalasi Listrik (General Regulation of Electrical installation) Lembaga Ilmu Pengetahuan Indonesia
SAMA	Scientific Apparatus Manufacturer's Association
SAE	Society of Automotive Engineers, USA
SII	Standar Industri Indonesia (Indonesian Industrial Standard) Pusat Standardisasi Industri, Departemen Perindustrian,
SLI	Standar Listrik Indonesia
SNI	Standar Nasional Indonesia
SMACNA	Sheet Metal and Air Conditioning Bidder's National Association, Inc.
SPLN	Standard of Perusahaan Umum Listrik Negara Pusat Penyelidikan Masalah Kelistrikan (LMK)
SSPC	Structural Steel Painting Council
TEMA	Tubular Exchanger Manufacturers Association, Inc.
UBC	Uniform Building Code of USA
UL	Underwriters Laboratories, Inc.

(Last page of Subsection 2.5)

SUBSECTION - 2.6 PERFORMANCE TESTING, GUARANTEES AND FIRST INSPECTION

2.6.1 GENERAL

The equipment and systems to be supplied under this Contract shall undergo tests for verification that the components satisfy all the performance requirements as specified. All equipment tests shall be conducted in accordance with the codes and standards specified in this Document. Any equipment performance test not covered by codes or standards in this Technical Requirement shall be subject to the review and approval by the Owner.

Performance of individual tests specified herein shall not relieve the Bidder of his responsibility for his equipment guarantees required by this Specification.

2.6.2 SCOPE

Bidder shall conduct performance test which shall include all individual items of equipment, complete system and subsystems covered under this contract.

Bidder shall provide all equipment and instrumentation required for testing. The Bidder shall conduct all the required mechanical, electrical and I&C, running and functional tests for all applicable items of equipment prior to shipment as well as at Site.

It is Bidder's responsibility to carry out all performance tests in this Contract. The equipment at Site shall be operated by the Owner's operators during the tests under the direction and supervision of the Bidder's representative.

At the end of warranty period of the plant or part thereof for which a taking over certificate has been issued, the Bidder shall conduct inspections on the plant or part thereof to ensure the Owner that the equipment is in a good condition prior to issuance of Final Acceptance Certificates.

EQUIPMENT	PERFORMANCE TEST CODE	TEST LOCATION
Performance Test Code on Overall Plant Performance	ASME PTC 46	Site
Steam Turbine Unit	ASME PTC 6	Site
Heat Recovery Steam Generator	ASME PTC 4.4	Site
All Centrifugal Pumps	ASME PTC 8.2	Shop
Generators and Electrical Rotating Machine	IEC 60034	Shop
Deaerator	ASME PTC 12.3	Site
Condenser	ASME PTC 12.2	Site
Heat Exchanger	ASME PTC 12.1	Site
Air Compressor System	ASME PTC 10	Shop
Generator and Electrical Rotating Machine	IEC 60034	Shop
Power Transformer	IEC 60076	Shop
Excitation	IEEE	Shop
Switchgears Equipment	IEC	Shop
Power Cables	IEC	Shop
Compressed Gas Insulated Bus (CGIB)	IEC	Shop & Site
Grounding System	IEEE 80,142 & 665	Site
Vibrations	ISO 10816	Shop & Site
Water Treatment Plant	*	Site
Waste Water treatment	*	Site
Chlorination Plant	*	Site
Desalination Plant	*	Site
Fire fighting equipment	*	Site

EQUIPMENT	PERFORMANCE TEST CODE	TEST LOCATION
Emergency diesel generator	*	Site

Note: *) Bidder shall propose codes or standards for Plant performance test to be approved by the Owner.

The Bidder shall inspect all major equipment at the end of the Warranty Period.

2.6.3 TEST PROCEDURES

Preliminary test procedures for all equipment shall be submitted at least six (6) months before the scheduled test date, respectively.

The Bidder shall submit detailed test procedures for all equipment specified in this Contract. The Bidder shall submit these procedures to the Owner at least two (2) month before the scheduled test date. These test procedures shall include but not be limited to:

- Details of all tests including performance tests either conducted in the manufacturer's works or at Site.
- Applicable codes and standards.
- Flow diagrams showing all points of measurement for both functional and performance tests together with logic diagrams showing the correct plant functioning requirements.
- Extent of test and permanent instrumentation to be used.
- Methods of calibration for test instruments, and accuracy of each instrument to be used.
- Elapsed time test program and testing activities.
- Manpower requirements and resources.
- Method of calculations including formulas and correction factors to be used.
- Test results.
- Format of test report including pro-forma schedules for the recording and presentation of test data.

2.6.4 EQUIPMENT PERFORMANCE

Performance tests at Site shall be conducted as required by **Part 3, Clause 13** and **Part 4 Section 4**, on the Add On block in the Bidder's supply in order to verify that the equipment satisfies all the design requirements, and to demonstrate that the performance and guarantees specified in the Contract have been met. The Bidder shall provide and install all necessary test points, instruments, fittings and other equipment and consumables except fuel, water and electricity required for conducting the performance tests.

Correction factors for variations of test conditions from the specified design conditions shall be as stated by the Bidder in his Bid Proposal, and agreed in the Contract.

The performance tests shall, among others, include the verification of following guarantees mentioned in **Part 5, Schedule 6**:

- Plant Guarantees for each block on Gross Power Output and Auxiliary Power Demand, refer also to **Clause 2.6.5** below
- Guarantees on Production Rates of Auxiliary Plants
- Guarantees on Noise Emission
- Guarantees on Equipment Vibrations

Further required guarantees on characteristics of individual items of equipment may be specified in the technical specifications concerned.

2.6.5 PLANT GUARANTEES

The Bidder / Bidder shall guarantee Gross Power Output and Auxiliary Power Demand as described below, refer also to **Part 4, Subsection 4.9**.

2.6.5.1 Guarantee Conditions:

The plant guarantees shall be based on the following ambient and operating conditions:

- Air dry bulb temperature 30°C
- Relative humidity 83%
- Barometer pressure 1013 mbar
- Seawater inlet temperature 32.5°C
- Rated power factor 0.80

- Frequency 50 Hz
- Existing GT parameters, refer to Part 5 Schedule 6

The Bidder shall furnish full details of correction (including correction curves) to be applied for departures from the above parameter required.

2.6.5.2 Gross Power Generation

The Bidder shall guarantee the Gross power generation according to Part 5, schedule 6 at maximum continuous rating (MCR) for :

- Muara Tawar Block 2 Add On Steam Turbine combined cycle block in the combined cycle mode
- Muara Tawar Block 3 and Block 4 Add On Steam Turbines combined cycle block in the combined cycle mode

The gross power output shall be measure by energy meter (kwh) or electric power (kw) at terminal generator output.with class accuracy as ASME PTC 4.6

2.6.5.3 Auxiliary Power Consumption

The Bidder shall guarantee the auxiliary power consumption corresponding to MCR of each add-on auxiliaries of combined cycle block.

The Bidder shall identify each power users included in the above guarantee for add-on auxiliaries of combined cycle block, separately.

The Bidder shall also identify and guarantee power consumption required for all common ancillary plants.

2.6.5.4 Correction Curve

The Bidder shall provide and guarantee the correction curve of power output in conjunction with the fluctuation of gas turbine exhaust parameter and shall be submitted together with Schedule 6 Plant Performance Guarantee.

2.6.5.5 Bonafide Tests

The Bidder shall perform bonafide test on partial load of combined cycle block with the condition of:

1. One (1) Gas Turbine in operation for each Block
2. Two (2) Gas Turbines in operation for Block 3 & 4
3. Efficiency for each HRSG
4. Steam Turbine Generator Heat Rate

2.6.5.6 Test Period and Test Condition

a. Test Period

Each test shall be carried out for two-hours period continuously at each load tested. The test period shall be selected at a time when the power system is stable and variation of the frequency and generator power factor is small.

ADD2
-IIA
-008

b. Test Conditions

- The HRSG shall be operated automatically and shall maintain each steam pressure and temperature at the specified value.
- All systems shall be operated automatically
- The generator load shall be constantly maintained at the required load (by the load limiter at partial load)
- Blow off and continuous blow down from HRSG shall be closed
- Make-up water inlet valve shall be closed
- Condenser back-wash shall not be carried out during the test
- Auxiliary steam use shall be limited for only plant operation

2.6.5.7 Instrumentation accuracy

The Bidder shall prepare the following instruments with the certifications of the authorized party. The Owner shall have the right to witness calibration of these instruments. Such instruments will be of precision class advised by selected codes and standards and will be calibrated inclusive of traceable certificate. The

achieved overall measurement uncertainty shall be equal or smaller than that as stated in ASME PTC 46.

- Fuel Gas Flow Meter Orifice type, Turbine meter type, or Ultrasonic type, accuracy $\pm 0.2\%$
- Feed water flow meter Orifice or Flow nozzle type, accuracy $\pm 0.2\%$
- Flow meter Positive displacement type, accuracy $\pm 0.2\%$
- Generator output meter Standard type accuracy $\pm 0.2\%$
- Auxiliary power meter Standard type accuracy $\pm 0.2\%$

2.6.6 TEST BEFORE SHIPMENT

Before any equipment is packed or dispatched from the Bidder's or Sub Bidder's works all relevant performance and other tests called for in this Contract shall have been successfully carried out in accordance with the Inspection and Test Program (ITP's) and appropriate codes and standards and to the satisfaction of the Owner, who shall have seen and approved the test results.

The Bidder shall supply six (6) bound copies of all certification, suitably indexed, to cover all performance tests of equipment.

2.6.7 TEST RECORDS

During all tests a complete and detailed log of observations and readings shall be made and certified copies submitted to the Owner.

The detailed test procedures, lists, logic diagrams and flow diagrams together with test results shall be correlated and presented as a single bound final test report to form a comprehensive set of test procedures and records of the tests conducted on the plant.

Six (6) bound copies of the Final Test Reports shall be supplied by the Bidder to the Owner.

2.6.8 GUARANTEE AND REJECTION

If any equipment is found unable to meet guaranteed performance stated in **Part 5 Schedule 6** or elsewhere in the specification, the Bidder shall repair and/or replace such defects at his own expenses, and shall re-verify guarantees.

The time period for repair will be decided by the Owner if it is beyond the specified completion date.

If the Bidder fails to repair and/or replace within the time period mentioned above, then the Owner has a right either to reject or to apply Liquidated Damages for performance, refer to Part 2, Clause 38 and Part 3 Clause 4.

2.6.9 FIRST INSPECTION

At the end of warranty period for each item or equipment, the Bidder shall perform visual inspection of the equipment as a condition for the Owner's issuance of the Final Acceptance Certificate.

Any defective work discovered during this inspection shall be remedied free of charge by the Bidder under the terms of the warranty of the Contract. An inspection procedure shall be submitted to the Owner for approval not later than two (2) months before the scheduled inspection. The inspection shall be conducted according to mutually agreed conditions.

An inspection report shall be submitted for each piece of equipment or system inspected. This report shall include, but not be limited to, the following:

- Status of equipment or system.
- Deviation from expected conditions.
- Remedial action required.
- Conclusions.
- Mutually agreed correction procedures.
- Photographs.

(Last page of Subsection 2.6)

SUBSECTION - 2.7 QUALITY ASSURANCE / QUALITY CONTROL (QA/QC)

2.7.1. STANDARD QUALITY ASSURANCE/QUALITY CONTROL

2.7.1.1 Bidder's Quality Assurance/Quality Control System Description

Bidder shall submit with his Bid Proposal, two (2) copies of his Quality Assurance Program and Quality Control Manual for review and acceptance.

The Quality Assurance Program shall meet the requirements of ISO 9001/2000 or equivalent, as applicable.

In all cases, the Quality Assurance Program must address the organization of the Bidder.

Bidder shall submit with his Bid Proposal, a preliminary Project Quality Plan including a list of Project Procedures, and Quality Control Documentation (Inspection and test Program) which apply to the Work and which he intends to submit to the Owner. This QA documentation will be reviewed during the bid evaluation and mutually agreed upon and will be included as part of the Contract requirements.

2.7.1.2 Detailed Quality Control Program

Two (2) copies of the detailed Quality Control Program for the Works shall be submitted to the Owner for approval not later than six (6) weeks after Award of Contract and at least four (4) weeks before the start of fabrication or installation which ever is earlier.

To maintain the equipment and other items in a corrosion-free state, the Bidder shall submit the Owner with instructions for storage and maintenance for his equipment and other items prior to installation, as well as for the period between the completion of the installation and the time that the equipment and other items are placed in service.

The Bidder shall provide a Site Quality Control program including requirements for Site Quality Control surveillance, hold points and in-progress inspection points.

2.7.1.3 Overall Inspection Test Program

Within three (3) months after Award of Contract or four (4) months prior to the start of fabrication whichever is earlier, Bidder shall submit to the Owner an Overall Inspection and Test Program (ITP) which shall include pertinent fabrication and inspection operations which will be of interest to the Owner relative to Quality Control. The Owner shall select one (1) or more of these points as a mandatory Customer Notification Point (CNP) Program for surveillance.

The Customer Notification Point Program shall include all mandatory hold points prior to shipment.

The Bidder will give the Owner at least ten (10) working days prior notice in writing of, the date on and the place at which any plant will be ready for testing. Unless the Owner shall advise the Bidder within seven (7) days of the date which the Bidder has stated in his notice the Bidder may proceed with the tests, which shall be deemed to have been made in the Owner's presence, and shall forthwith forward to the Owner duly certified copies of the test readings.

2.7.1.4 Sub Bidder's QA/OC System and Procedures

The Bidder shall be responsible for the review, comment and acceptance of the Quality Control System, Quality Control and Fabrication procedures of his subBidders. In addition, Bidder shall be responsible for his subBidder's work. The Bidder shall submit to the Owner a list of any or all of the work to be sub-contracted.

2.7.1.5 Documentation Required

Three (3) copies of all appropriate documentation, as specified by the Technical Specifications and as required by applicable codes and standards shall be submitted to the Owner for review and acceptability prior to, or concurrent with, shipment or as otherwise agreed to by special arrangement with the Owner.

2.7.1.6 Contract Non-conformances

Non-conformances to the Contract requirements shall be promptly submitted to the Owner in writing for resolution. After submittal of non-conformances, the Bidder may proceed with any work at his own risk.

2.7.1.7 Radiographs

The following procedures shall be followed in the radiography of the applicable equipment and materials.

All shop radiographs, with appropriate reports are available on the shop site for review and will be submitted to the Owner with the End of Manufacturing Report (EOMR). The double film technique shall be employed if the Bidder consider it necessary, otherwise a single film technique is sufficient. All radiographs shall become the property of the Owner. All radiographs, not properly identified, will be rejected. Packaging and shipment of radiographs shall be in accordance with Clause 2.7.2 herein.

All field radiographs, with appropriate reports are available on the site for review and will be submitted to the Owner within ten (10) days after completion of Radiography. All radiographs shall become the property of the Owner. All radiographs, not properly identified, shall be rejected. Packaging and shipment of radiographs shall be in accordance with **Clause 2.7.2** herein.

2.7.1.8 Legibility of Documentation

All documentation shall be clear, legible, and of suitable quality for storage for the life of the Plant. Radiographs will not be microfilmed.

2.7.1.9 Honoring of Invoices

Invoices for equipment received on-Site will not be honored until all documentation has been received and complies with Contract documents.

2.7.1.10 Completion Letter

After delivery of all documentation required as specified the Bidder shall submit a letter to the Owner listing all of the documentation delivered and stating that all requirements of the Specification have been fulfilled.

2.7.2. STANDARD PROCEDURE FOR PACKAGING AND SHIPMENT OF RADIOGRAPHS

2.7.2.1 General

This procedure covers the packaging and shipment requirements for Bidder's submittal for radiographs to the Owner (for shop radiographs) for review and comment when required by the Specification.

Each radiographic film package shall be identified as follows:

- a. Owner
- b. Plant and Unit No.
- c. Specification No. and Latest Amendments
- d. Component Name and Identification No.
- e. Heat No., Piece No., Register No., Serial No. (as applicable)
- f. Number of films contained in the package
- g. Weld or Seam No. (if applicable)

2.7.2.2 Packaging of Radiographs

Preparation for packaging: The radiograph shall be assembled in accordance with the equipment supplier's standard practice for handling and filing radiographs.

Inner carton: The assembled radiographs shall be inserted into a heavy fiberboard or cardboard carton, and all openings sealed. Radiographs must not be folded.

Outer box: The inner carton shall be packed in a wooden box constructed to protect the radiographs from shipping damage and the elements.

2.7.2.3 Shipments of Radiographs

Shop Radiographs

- a. Radiographs shall be shipped with a transmittal letter to the Owner via a carrier that can trace its shipment.
- b. The radiographs shall be insured for a nominal amount.

- c. Bidder shall request a return receipt from carrier to verify shipment was received.

Field Radiographs

- a. An original and a copy of the transmittal letter and all radiographs shall be sent to the Owner at the Site.
- b. Radiographs shall be packaged as required by Clause 2.7.2.2 above, except outer box, and taken to the Owner on Site.

2.7.3. SHOP INSPECTIONS

2.7.3.1 Shop Inspection and Test Program / Inspection and Test Time Schedule

The Bidder shall submit a coordinated detailed shop inspection test program (ITP) / Inspection and Test Time Schedule covering all quality control procedures required including at least the following information in a format to be approved by the Owner:

- Component description
- Individual inspection or test item
- Verifying documents type
- Who witnesses (Bidder, sub vendor, Owner, etc.)
- Location of inspection or test
- Estimated test date (month, year)
- Test record submission (at test, at shipment, or none [stored by Bidder])
- Remarks (any clarifications, etc.)

The plan shall show all Bidder or Sub vendor items shop inspections and tests at all stages of manufacture, including raw materials, such a steel plat from the mill, etc.

The Bidder may provide other relevant information in addition to the above and shall submit his proposed format for the inspection and test plan for Owner approval within two (2) months of Contract signing, and the completed plan within four (4) months of Contract signing. Separate plan shall be submitted for both domestic and overseas manufacturing activities.

2.7.3.2 Shop Inspection Schedule

The Bidder shall also provide a coordinated detailed inspection schedule covering all customer notification points update regularly for changes in schedule, giving the expected week a specific inspection is expected to take place.

2.7.3.3 Shop Inspection Procedures

For all shop manufactured items, shop inspection procedures shall be submitted for the Owner's approval not later than one (1) month prior to the estimated inspection date. The Owner shall give final approval before relevant actual inspections are scheduled.

2.7.3.4 Shop Inspection Notification

The Bidder shall notify the Owner of the scheduling of shop inspection activities to be witnessed by the Owner at least twenty one (21) calendar days before the scheduled activity for overseas manufactured items and seven (10) calendar days before for domestic manufactured items. The notification letter shall be submitted in accordance with project correspondence requirements and in parallel by fax. The notification letter shall include as a minimum:

- Equipment or material to be inspected
- Location of inspection
- Contact person at factory
- Date of inspection
- Inspection procedure document number
- Approved design drawing number(s)

It is important that this information be submitted completely as specified in order to provide for efficient, timely and effective the Owner's monitoring of the inspection process and attendance at key inspections points and tests.

2.7.3.5 Document Approval Status

As required in Contract document **Part 4 subsection 2.2**, manufacturing shall be performed based only on approved project drawings. Inspections shall be

based only on final approved project drawings. Bidder shall not schedule such inspections or ship any material or equipment prior to receiving final approval.

(Last page of Subsection 2.7)

SUBSECTION - 2.8 PLANT IDENTIFICATION

2.8.1. GENERAL

This general technical requirement defines the requirements, conventions and guide lines for assigning numbers and titles to the equipment that is to be numbered.

The Bidder shall prepare comprehensive Plant Identification Schedule showing the name and number of each item of Plant and its respective arrangement drawing number, and add any additional items necessary to fully identify the Plant.

This schedule shall be subject to the Owner's approval.

All equipment in the designated system shall be assigned with equipment numbers. A single, comprehensive equipment numbering system shall be applied uniformly to all plant equipment.

The same equipment number shall not be applied to more than one (1) equipment.

Equipment number shall be depicted on the P&IDs, physical piping and general arrangement drawings, specification, system or functional diagrams, electrical drawings, schedule, civil and structural drawings, etc.

Plant identification and equipment number shall be derived from European Identifications system for Power Plant (KKS).

2.8.2. NUMBERING SYSTEM

The direction of identification number or letter advancement shall be to Site North and to Site West to the Compass.

This means that the equipment or plant item located on the Site North side has a higher identification number than the equipment located on the South side, and equipment located on the Site West side has a higher number than one which located on the Site East side.

The direction of identification number of equipment of plant items that is installed in the vertical arrangement, advancement shall be to the upper side.

2.8.3. NAMEPLATES AND LABELS

The Bidder shall supply all labels, nameplates, instruction and warning plates necessary for the identification and safe operation of the Plant and all inscriptions shall in the English language and the size of the lettering shall be to the approval of the Owner. Main nameplates shall in the Indonesian language.

The inscriptions on all plates shall be engraved with black-filled lettering either at Site or in the manufacturer's works after the equipment designations has been finalized.

Where applicable plates and labels shall provide a white background to the inscriptions. Approved, temporary labeling is to be provided until such time that this rationalization is complete.

Inscriptions shall be to the approval of the Owner and where applicable shall include the plant nomenclature in addition to the alphanumeric plant designation code as described above.

The plates and labels shall be non-deteriorating and non-warping under Site conditions. Stainless steel is to be used on outdoor equipment. Traffolyte, or an approved equivalent, may be used for individual component labels inside panels and desks.

All labels, nameplates, instruction and warning plates shall be securely fixed to items of plant and equipment with stainless steel rivets, plated self tapping screws or other approved means. The use of adhesives will not be permitted.

Warning plates shall be manufactured from stainless steel with a matt or satin finish, engraved with red lettering and located in a position that shall afford maximum personnel safety.

All equipment within panels and desks shall be individually identified by satin or matt finish stainless steel labels or laminated plastic labels where approved.

Each circuit breaker panel, control panel, relay panel and the like shall have a circuit designation label mounted at both front and rear. Corridor type panels

shall additionally have circuit designation labels within the panels. The function of each relay, control, indicating and alarms device; fuse, etc. shall be separated.

All unfired pressure vessels and lifting tackle shall have permanently attached to them or stamped on them, in a conspicuous position, a rating plate giving such details as the design code, design parameter, test parameter, date of test, etc. to the approval of the Owner. Plate shall be installed on vessel's insulation so as to be visible after insulation is applied.

Pipe work systems shall be identified in accordance with the Owner's standard color identification system.

Each valve shall be fitted with a stainless steel nameplate indicating the valve service and reference number in accordance with the valve schedules as approved by the Owner. The color of the valve nameplate engraving shall correspond to the service of the pipe work in accordance with the requirements of appropriate standards.

Where possible valve nameplates shall be circular and fitted under the hand-wheel captive nut. On check valves and small valves the Bidder may provide rectangular nameplates fitted to brackets on the valve or attached to a wall or steelwork in a convenient position adjacent to the valve.

Generally the plate material shall be stainless steel, for the water treatment plant plastic plates shall be used.

Size of plate and lettering shall be to the approval of the Owner.

(Last page of Subsection 2.8)

SUBSECTION - 2.9 GENERAL NOISE CRITERIA

2.9.1 GENERAL

These criteria define permissible noise levels at specified locations in and around the plant during the plant operation.

The following criteria are applicable to all the equipment furnished under this Contract whether referenced specifically elsewhere in the Terms of Reference and Technical Specification Document or not:

- Hearing conversation to protect plant personnel from hearing loss and personnel discomfort.
- Annoyance to community - to prevent annoying adjacent communities.
- Background noise level - to permit good voice communication.

2.9.2 CODES AND STANDARDS

The following codes and standards shall be applied in designed the noise levels of equipment and other accessories and buildings to be supplied and installed under this Contract:

- PLN's Standards (SPLN 46-1and SPLN 46-2)
- U.S Department of Labor, Occupational Safety and Health Administration (OSHA).
- ANSI SI-13 Methods for the Measurement of Sound Pressure Levels and appropriate standards approved by the Owner.

2.9.3 GENERAL DETAILED REQUIREMENTS

Bidder shall guarantee that the operating equipment will not produce a sound pressure level greater than 85 dB(A) when measured one (1) meter from any point on any item of equipment or enclosure and 1.2 meter above floor including valves during steady and normal operation. This requirement applies to the running plant taking into account the noise effect from all the running equipment within this Plant.

During periods of abnormal operation e.g. turbine trip, ST bypass operation, steam blowing, silencers downstream of safety valves etc. a noise pressure power level of 110 dB(A) in 1 m distance shall not be exceeded.

Permissible noise levels inside building during plant operation the following limits shall not be exceeded with the relevant room or building in its normal of use, and hence will generally apply with windows and doors shut and with air-conditioning plant in operation. The Bidder shall therefore pay due attention in his design to noise contributed by operation of the Contract works outside the room, the noise arising from the air conditioning of the room, and noise generated by Contract works positioned in the room.

- (a) New control rooms and conference rooms for Block 3 and Block 4: 45 dB (A), (combined control room and conference room for Block 1 and Block 2 is existing).
- (b) Inside all buildings with extended occupancy by operation and maintenance staff, unless otherwise specified 60 dB (A)
- (c) **If any** Interior of all buildings and enclosures unless otherwise specified or unless those enclosures are themselves provided for noise protection of equipment 80 dB (A).

ADD2
-IIA
-007

The maximum noise level measured at the boundary of the power plant (approx. at the power plant fence) shall be 60 dB(A) maximum between 6 a.m. and 10 p.m. and 50 dB(A) maximum between 10 p.m. and 6 a.m.

The maximum level at the nearest residential areas shall be 60 dB(A) between 6 a.m. and 10 p.m. and 50 dB(A) between 10 p.m. and 6 a.m.

The noise level measurement shall be in accordance with ANSI SI-13 and / or SPLN 46.

(End of Subsection 2.9)

SUBSECTION - 2.10 INSTRUCTION MANUALS

2.10.1. GENERAL

The Bidder shall provide sets of well structured, comprehensive, and coordinated manuals to fully describe all aspects of design, operation and maintenance of all equipment agreed under the Contract including tools and testing equipment. The format of the document will be agreed with the Owner before the Bidder proceeds to assemble the draft manuals.

Three (3) hard copies and five (5) DVDs of the first draft of the manuals shall be submitted to the Owner for review six (6) months before start cold commissioning of the plant.

Three (3) hard copies and five (5) DVDs of the final manual shall be submitted to the Owner one (1) month prior to the commencement of commissioning of the plant.

Each manual shall be arranged in three (3) separate parts, namely design (technical) documentation, operation and maintenance including spare parts, and digital systems. As a minimum a complete self standing manual shall be submitted for each equipment.

The layout of the three (3) parts should, as far as possible, be consistent. System, plant and equipment shall be described in the same order in each section.

Each section shall be divided into as many volumes as necessary for convenient handling and reference.

All plant and equipment identification shall be by means of the description and numbering system specified in Part 4, Subsection 2.8.

The manual format shall be on standard metric A4 sheets. Drawings and schedules, which are to be bound into the manual, shall also be A4 or A3 folded to A4 sizes.

The manuals shall include instructions only for the actual equipment supplied and not for alternative or optional equipment. If generic documents are included the non-applicable portion shall be crossed out.

All documents, illustrations, specification, operating and maintenance instructions and parts shall be in the English language, and the units in metric system.

Each individual major item of equipment specified shall have its separate instruction manual. All information submitted after Contract Award should be incorporated into the instruction manuals.

Parts shall be identified by the original manufacturer's part number as well as by Bidder's identification number..

Bidder shall also be held responsible for supplying additional information or replacing information to keep the manual complete, accurate and update if field inspections of equipment indicate omissions or inaccuracy of the manuals.

All manuals will be submitted complete in durable binders expected to last over multiple years and which facilitate addition or removal of information.

2.10.2. DESIGN MANUALS

The volume(s) comprising the design part to the manual should be laid out as follows:

Preliminary pages including index, preface and an amendment record sheet and illustration of the complete equipment

General description of the system including:

- System diagrams and blocks diagram showing sub-systems and interconnection with other systems.
- System description, including design basis, function, location, and modes of operation.

Design data including performance curves, heat balance diagrams, materials specification and running clearances and settings, limitations, etc.

Instrument and valve list itemizing function, type, number, range and alarms.

2.10.3. OPERATIONS MANUALS

The volume (s) comprising the operation part of the manual shall be laid as follows:

- Preliminary pages including index and amendment record sheets
- Equipment identification by plant name, unit number and by function name
- Recommended procedure for operating and instruction for commissioning, start-up, normal operation, shut-down, standby, emergency (including loss of normal power) and fault conditions, and other information such as normal range of system variables, operating limits and hazards conditions for alt equipment and systems
- General Arrangement Drawings
- Final performance and design data sheets and performance curves for all equipment
- Testing and checking requirements

2.10.4. MAINTENANCE MANUALS

The volume(s) comprising the maintenance part of the manual should be laid out as follows:

- Preliminary pages including index and amendments record sheets
- List of names and addresses of liaison offices for all equipment
- Preventive Maintenance and Inspection Schedule for all equipment covered by this Specification
- Lubrication Schedule showing requirements, and specifications for all equipment covered by this Specification and showing equivalent lubricants available in Indonesia
- Recommended procedure for assembly and dismantling with associated tests and check-ups prior to returning equipment to service
- Detailed construction, assembly drawing to complement assembly procedures mentioned above, including part list and numbers for replacement ordering
- Setting and running clearances and tolerances
- Methods for trouble shooting
- Cleaning and conservation procedures
- Instrumentation calibration requirements and procedures

The listing of all replacement parts/spare parts list identifying the following:

- Name of spare part
- Identification number
- Manufacturers drawing number
- Installed quantities and spare quantities
- Outline sketch diagram
- Materials
- Required period delivery.

Drawings showing space provided or equipment maintenance for all equipment and any fixed facilities for maintenance such as trolley beams, etc.

2.10.5. DIGITAL SYSTEMS MANUAL

For digital control systems the Bidder shall also submit a manual describing the digital system, system generation, software and hardware to include the following as a minimum:

- The manual shall include functional descriptions including intended use, capabilities, limitations, control logic diagrams, control set points and correction curves, etc.
- The manual shall contain the procedure required to generate the system software.
- The software portion of the instruction manual shall describe the software provided with the system including the self-diagnostics, operating system, the utility software and the application software. This shall also include reference, user and programming manual for all software supplied.
- The hardware portion of the manual shall contain physical descriptions of system hardware, functional characteristics, capabilities, specifications, data sheets, and special characteristics. Assemblies and subassemblies shall have unique part numbers that shall be referenced in the documentation including in the manuals. This shall also include theory of operation and general overview of the system hardware furnished.

(Last page of Subsection 2.10)

SUBSECTION - 2.11 SPARE PARTS, TOOLS, TEST EQUIPMENT AND CONSUMABLES

2.11.1. SPARE PARTS

The Bidder shall ensure that sufficient spare parts are available for his use during test and commissioning of the Plant. Such spare parts shall be in addition

to any spares provided under the Contract as Mandatory Spare Parts and Recommended Spare Parts.

The price for mandatory spare parts shall be included in the bid price.

Should the Bidder require any spare parts that may be available in the Owner's stores, these may be issued to the Bidder at the Owner's option. In such a case the Bidder shall immediately arrange for the spares to be replaced with new items to complete the Owner's holding.

The Bidder shall include in his Bid proposal a list of those spare parts and quantities he recommends should be supplied for two (2) years continuous operation, having regard for the remoteness of the Site from suitable sources of such spares.

The price for recommended spare parts shall be given in **Part 5 Schedule 2.3** and shall be fixed not subject to escalation for period of two (2) years after taking over of the Plant. The price and quantities of mandatory spare parts shall be listed in **Part 5 Schedule 1.4**.

The Bidder shall give assurance in his Bid proposal that spare parts will be reasonably available throughout the life of the Plant.

The quotation shall include the unit price, quantity, description, catalog number, part number, drawing references, etc. for each spare part to completely identify the items and the equipment component for which it is recommended.

The Owner may order all or any of the spare parts at his own discretion. Those ordered are to be delivered to the Owner on the Site not later than the date of issue of the Final Acceptance Certificate for the respective item of plant or equipment.

The Bidder is required to submit the schedule of spare parts listing the quantity of each spare recommended to be held in stock, individual prices and normal delivery period.

All spare supplied shall be strictly interchangeable with the parts which they are intended to replace and shall be tested and packed for long storage under the climatic condition prevailing at Site.

Each spare shall be clearly marked with the description, purpose and plant designation code.

Any spare having a limited shelf life or requiring special storage condition shall be separately identified.

If the Bidder's practice is to assign his part numbers to parts purchased from other manufacturers, a part number list shall be furnished which provides a cross-reference between the Bidder's part number and the part of the original supplier.

Spare parts shall be shipped separately from the balance of the Contract and be assigned to the Owner at Site. They will then be stored by the Owner.

All cases, containers, and packages are liable to be opened for examination by the Owner and all such opening and subsequent repacking shall be at the expense of the Bidder.

Bidder shall include within his spare parts list those spares necessary to enable the Owner to carry out routine maintenance on every instrument as recommended by the manufacturer in his catalogue.

The Bidder shall also include these spares necessary to maintain the special test and calibration equipment to be supplied under the Contract.

2.11.2. TOOLS AND TEST EQUIPMENT

Any special tools and test equipment and standard tools and test equipment required to adjust, dismantle, test or maintain any equipment furnished by the Bidder shall be provided by the Bidder, except special tool required for inspection of the gas turbine.

A complete list of tools and test equipment shall be included in the Bid proposal.

Tools and test equipment shall be supplied separately for section of the Works as follows:

- Heat Recovery Steam Generator unit and associated auxiliary equipment
- Steam turbine generator unit and associated auxiliary equipment
- Instrumentation and control system equipment
- HV, MV and LV electrical equipment

- Each ancillary plant

When tools or test equipments are common to more than one (1) section, then sets shall be provided for each section.

The equipment provided shall comprise, but not be limited to, the following items for each section as applicable:

- a. Two (2) sets of all spanners, wrenches, screwdrivers, pullers, extractors, torque wrenches, etc to suit every item of plant supplied under this Contract.
- b. One (1) set of lifting slings, beams, up-ending devices, eye-bolts, chain blocks, etc. including all equipment required to remove and replace any rotors, including temporary support structures.

All such equipment shall be tested prior to delivery and supplied with duplicate copies of the test certification.

- c. One (1) set of equipment required for withdrawing couplings, bearings, impellers, glands, etc.
- d. One (1) set of any equipment required for removing replacing and plugging each type and size of heat exchanger tubes including tube expanding equipment.
- e. Valve lapping equipment.
- f. One (1) set steel frame complete with lifting arrangements for each different type of turbine rotor, suitable for safe and convenient transport of the respective rotor fully bladed from to the power plant.
- g. One (1) set of any special tools required during maintenance and cleaning of the Works, including bridge gauges, alignment jigs, etc.
- h. Two (2) set of bolt tightening equipment for turbine.
- i. One (1) set of taps and dies in fitted boxes to cover the whole range of plant.
- j. One (1) complete set of Instrumentation and Control system testing and maintenance tools including card extenders, test connectors on performance cable incorporating special plug or sockets as appropriate, special testing, calibrating, diagnostic equipment (including CRT, consoles and printer), measuring devices, etc. as particularly required by

the various equipment supplied in this Contract in accordance with each manufacturer's recommendation.

- k. One (1) set of racks, power supplies and back panel connections for the testing of control modules.
- l. One (1) set of any special tools and equipment required for the testing and maintenance of electrical and electronic components.

Tools and test equipment shall be new and of good quality.

Tools and test equipment shall be neatly arranged in a box and shall be shipped to the job in suitable, separate container clearly marked with the name of the equipment of plant for which the tools and test equipment are intended.

In general these tools and test equipment shall not be used during erection and commissioning.

The Bidder shall furnish a complete set of tools for erection and commissioning at his own expenses.

However, if major special tools and test equipment become necessary during erection, the Owner will, hand over the tools and test equipment to the attention and responsibility of the erection supervisor.

The tools and test equipments shall be returned in good condition or shall be replaced at the Bidder's expenses.

2.11.3. CONSUMABLE PARTS AND MATERIALS

The Bidder shall furnish a sufficient quantity of required consumable parts and materials such as gaskets, seals, lubricants, hand-hole covers, computer papers, recorder charts, ink, chemicals, **(chemical for operation of HRSG, Desalination plant, Water treatment Plant, Waste water plant, Electrochlorination, fuses, indicating and lightning lamps, filters, battery acid etc.) cleaning agents, paper roll, ink, oil, grease** etc . during erection, testing and commissioning up to the respective COD of that Block and during

ADD2
-IIA
-005

the warranty period. These consumable parts include initial charge of all lubricating oil and greases.

ADD1-
IIA
-018

These consumables shall not be quoted as spare parts furnished under this Contract.

The Bidder shall submit to the Owner within six (6) months after Contract sign, a schedule of consumable items, listing the items and the quantities required per month for operation of the Plant.

This schedule shall contain sufficient information to enable the Owner to put on order the consumable items listed.

A list of consumables shall be shown in **Part 5, Schedule 1.6 “Schedule of Consumables”**. The Bidder shall supply and maintain as a minimum, the quantities shown on the list.

The Bidder shall provide, without additional cost, whatever consumable materials are required over and above those shown on in the list during the execution of the Contract.

At the end of the Contract, the Bidder shall hand over to the Owner any of the material shown on the list, which remains unused.

All hardware, which has a life of less than one (1) year shall be replaced by new hardware by the Bidder at the time of taking over and shall be tested immediately prior to taking-over.

(Last page of Subsection 2.11)

SUBSECTION - 2.12 PACKAGING INSTRUCTIONS AND STORAGE

2.12.1 GENERAL

The Bidder shall package all equipment and materials furnished by him in such a manner to ensure protection against damage and deterioration during shipment and storage which includes local transportation, air and ocean freight, and Site storage. Procedure and details shall be submitted to the Owner for review prior to start of shipment.

The Bidder shall provide necessary storage facilities such as warehouse, protective covering, etc. for the purpose of proper Site storage. Storage procedures shall be submitted to the Owner, and such kind of storage shall be accomplished after Owner's approval.

All of the packages are liable to be opened for examination by the Customs Authority, or the Owner and all such opening and subsequent re-packing shall be at the expense of the Bidder. The Bidder shall retain responsibility for the prevention of pilferage, proper repacking and storage.

2.12.2 PACKAGING INSTRUCTIONS

The Bidder shall provide marks on all containers, cases, crates and packages in accordance with **Part 3, Clause 5.3** of this Document, for shipment and clearance purposes. Each package shall contain a complete detailed list of all the items therein.

All equipment shall be identified at the shop by means of a metal tag or other permanent material securely attached to the equipment. Tags shall remain on the units after erection, as a permanent record.

All random materials shall be tagged for identification.

All spare parts shall be packaged separately and be crated in a manner suitable for permanent storage.

All equipment including electric cabinets and parts shall be packaged for extended storage.

Equipment for different units shall be properly identified.

Tools shall be shipped to Site in separate containers clearly marked with the name of the equipment for which they are intended.

Small items shall be packed in sealed transparent plastic bags with desiccator packs as necessary.

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter, and shall be painted in accordance with the must protection requirements. The color of the prime coat shall be the Bidder's standard. No paint having an asphaltic base shall be used. Rust inhibiting solutions used internally shall be easily removable. Bidder shall ship a sufficient quantity of paint for touching up where prime coat has been damaged during shipment and erection.

The Bidder shall provide protection for all equipment before shipment as follows:

- All openings in equipment furnished by Bidder shall be securely plugged, capped or otherwise blanked off, sealed with tape and suitably protected against damage, and entry of foreign materials and moisture. For shop fabricated equipment this shall be done as soon as possible after shop cleaning.
- Weld ends on all valves, fittings, pipes, nozzles, etc. shall be capped and sealed with suitable, firmly attached protectors. Butt weld ends and ferrous materials shall be coated with anti-corrosion paint prior to capping, back to a ring 50 mm from the end.
- Protectors for bevelled ends designed for backing ring or consumable insert shall have a plywood or hardboard liner disc held securely against the bevelled end. Protectors shall not be welded to weld end.
- All flange facings, bolt holes and other machined surfaces of ferrous materials (except butt-weld end preparation) shall be coated with a suitable removable antirust compound.
- All flanged connections and loose flanges shall be provided with suitable full face flange protectors bolted in place and sealed.
- All machined surfaces shall be protected with wooden blocks or similar means and reinforced externally with metal bands or plates.

- All protectors for openings and all braces, brackets, spacers, ties, bindings and other shipping, packaging and packing materials and appurtenances used for protection in shipping, storing and handling of nonferrous piping and materials shall be of such design, type and/or arrangements as to prohibit contact between ferrous and nonferrous metals.
- Each section of pipe for lube oil systems and other systems, if specified in the Specification, shall be coated properly and scaled.
- All wheel bearings surface shall be properly protected. Shaft bearing surfaces shall be properly protected in accordance with the submitted procedures.
- Each section of equipment to be shipped in sections shall be reinforced to minimize distortion during shipping and handling. Disconnected piping and wiring shall be properly supported and braced for shipment.
- All moving parts requiring lubrication shall be greased in the shop before packing.
- All materials and equipment shall be packed for storage with adequate and suitable precautions to prevent any damage or deterioration in tropical and coastal environmental conditions.
- Materials susceptible to deterioration from climatic conditions shall not be used for the Plant unless otherwise approved by the Owner. These materials, if used for the Plant, shall be permanently protected from deterioration under the climatic conditions or, with the approval of the Owner, be coated with suitable surface treatment to protect the plant and equipment for a minimum period of twelve (12) months. The Bidder shall state the period of protection and specify treatment procedures and the recommended intervals between retreatment.

Prior to shipment, surfaces of wiring and all other parts susceptible in any way to moisture absorption or fungus attack shall be treated with fungicidal varnish.

The Bidder shall follow additional procedures as specified elsewhere in this Bid Document or as required in the judgment of the Bidder.

2.12.3 RECEIVING

Observations for unusual conditions encountered during transit shall be made by the Bidder or his representative to evidence the following:

- Tie down failure
- Rough handling
- Environmental damage
- Excessive exposure
- Fire or high temperature damage

Receiving Records

The Bidder shall assign qualified personnel to maintain control of the receipt, protection, and dispersment of the Owner's equipment and material. The Bidder shall give notice to the Owner upon receiving date. The Owner will be present at all equipment and material delivery inspections. A record shall be made of all equipment and material that is received, accepted or rejected. The record shall be as agreed upon with the Owner's Representative at Site.

Shortages

Upon receipt of shipments of equipment and material, quantities shall be checked against the shipping invoices. Shortages shall be immediately reported to the Owner's Representative at Site.

Marking

Received material in temporary storage shall be clearly marked, so that it can be readily located when required for installation.

Schedule Commitments

The Bidder shall be aware of the fact that the shortage or un-repairable damage to the equipment or material may cause delays in the project schedule.

Storage of Damaged Equipment

The Bidder shall be responsible for further damage to the rejected equipment and shall also store the equipment in a suitable area which will prevent further damage.

2.12.4 STORAGE

The Bidder shall provide detailed information for short term and long term storage and maintenance of his equipment prior to installation as well as for the period between completion of installation and the time that the equipment is placed in service.

Each type of preservative used by the Bidder shall be identified as to quality, life expectancy and type. Complete information shall be submitted to the Owner.

The Bidder shall load and unload the equipment and materials at the storage area and provide the maintenance of the storage and warehouse as required at his expense.

The Bidder shall supply storage work material such as dunnage, sheet cover, nitrogen gas, silica gel, touch-up paint, etc.

The Bidder shall provide necessary equipment for inspection and safeguarding of stored equipment and material and shall perform periodical inspection and repair work of stored equipments as required to maintain the equipment in warrantable condition.

The Bidder shall provide all manpower necessary for storage and removal at his expense. This shall include but not be limited to store clerks, inspectors, guards, handlers, etc.

Management of storage shall be provided by the Bidder. This shall include but not be limited to organizations, storage control, check sheets for storage condition, inspection records and security.

All storage areas shall be well drained while indoor storage areas shall also be ventilated.

Materials and equipment shall be stored above grade.

The equipment stored outdoors shall be covered with waterproof heavy duty canvas or plastic sheets. Storage covers whether canvas or plastic shall be fire retardant. The sheets shall be tightened to the equipment itself or its container with the rope passing through eyelets along the edge of the sheet, to prevent the sheet from being blown off by strong winds. After rains the Bidder shall temporarily remove the sheet to dry out any dew formed underneath the sheets.

Upon the arrival of the packages at Site the Bidder shall inspect and check each item of equipment as soon as possible. If any damage or deterioration is found, the Bidder shall take adequate and immediate remedial action. The Owner will also check the arrival of goods at Site and the Bidder shall facilitate Owner's inspection of the goods.

The Bidder shall provide weatherproof store house for instrumentation and electrical equipments. Other equipments to be installed indoor shall be stored properly and protected from damage, deterioration or contamination.

2.12.5 SPECIAL PROTECTIVE MEASURES

Certain items, by their nature or size, may not be stored in an area which provides the required level of protection and therefore the Bidder shall provide special measures.

Terminal bushings for high voltage transformers and high voltage circuit breakers may be stored outdoors in accordance with the manufacturer's instructions. Since bushings may be easily damaged, they shall be well protected and stored in a location out of the way of normal activity and traffic.

2.12.6 HEATER

Outdoor power transformer enclosures for tap-changers and cable terminal cabinets require internal protection for condensation, as do circuit breaker mechanism housing, outdoor switchgear, larger motor windings, etc.

These may be provided with electrical heaters. The Bidder shall provide temporary power and an external pilot light to monitor the supply feed at each item.

If the required heaters are not provided by the manufacturer or are incapable of preventing condensation during severe conditions, adequate heaters shall be provided, monitored and energized as above by the Bidder.

2.12.7 CABLE ON REELS

Both ends of electrical cable on reels shall be regularly inspected by the Bidder who shall ensure they are well sealed such as with several wraps of electrical tape before and during outdoor storage.

Protection against sun ageing, for cable on reels stored outdoors, shall be provided by a protective covering around the outer layers and furnished by the Bidder if not already provided by the cable manufacturer.

2.12.8 INSTALLED EQUIPMENT AND MATERIAL

All installed equipment and materials shall be protected from deterioration, condensation, physical damage and vandalism until the Work under the Contract is accepted in its entirety by the Owner.

The Bidder shall ensure that installed items are protected with the same level of protection accorded them while in storage.

Heaters shall be energized with temporary power and provided with external pilot lights to monitor the feeds. Dust contamination shall be avoided by suitable temporary coverings, with provisions for re-sealing on control panels and similar items.

Protection shall be provided by the Bidder for electrical cables in trays, ducts, risers and other electrical equipment when and where required, by means of fireproof covers or barriers to prevent damage from weld splatters, fires, water damage, falling debris, moving equipment, etc.

(Last page of Subsection 2.12)

SUBSECTION - 2.13 TRAINING

2.13.1 GENERAL

The objective of the Training Program is to train the Owner's operating staff (including manager level) in the designing, operation and maintenance of all equipment furnished under this Contract.

The Bidder shall provide training for the Owner's operating and maintenance personnel in the operation and maintenance of the equipment furnished by the Bidder.

The training shall include instruction at EPC engineering office and at Site. The training for operation and maintenance personnel should concentrate on the main equipment.

The Bidder shall ensure that the number of on site trainees and duration of training shall be sufficient to meet the requirements specified in herein.

Training Duration during Design Stage at the Bidder Offices

Description	Person	Weeks
Mechanical	4	2
Electrical	3	2
I & C	3	2
Civil	2	2
Quality Control (Q/C)	2	2

2.13.2 DEFINITIONS

2.13.2.1 Overview

Overview means an explanation of the principles of operation, method of manufacture, materials of construction, operational characteristics and an explanation of probable problems and their causes and remedies.

2.13.2.2 Maintenance

Maintenance means instruction in the detailed maintenance procedures required to keep the equipment in good operating condition

2.13.2.3 Troubleshoot

Troubleshoot means detailed instruction in diagnosing problems and is applied in this context primarily to instrumentation and control.

2.13.3 DETAILED REQUIREMENTS

The Bidder shall include in his Bid Proposal the proposed number of trainees, the duration of training and the type of training to be conducted by the Bidder at the manufacturer's facilities abroad. Number of trainees noted above are estimated only for bidding purposes.

The number of man months allowed for training may be increased, reduced or allocated to a larger or smaller number of trainees for a different time duration. Adjustments for under run or overrun of total man months will be by deduction or addition at the relevant adjustment rates in Part 5, Schedule 1.3. The rates for training abroad shall include for the cost of insuring trainees while undergoing works training against:

- Accidental death to a value of US\$ 200,000
- 100% of medical, dental and hospitalization expenses.

The cost of the trainees travel and living expenses including medical insurance, training facility etc. when assigned to a manufacturer's works shall be borne by the Bidder.

Training of twenty (20) trainees at Site will begin from, commencement of assembly of the Plant through Full Commercial Load. During this site training period two (2) full time instructors shall be on site continuously. The Bidder shall provide for instructors of the discipline required by the Owner. The maximum training period by each instructor shall be six (6) months cumulative.

The Bidder shall provide, from date of Full Commercial Load through to the end of Warranty Period, two (2) mechanical engineers and one (1) I&C engineer to provide supervision of and advice to the Owner's operation and maintenance personnel. These engineers shall have been qualified and approved by the

equipment manufacturers. Full details of the duties of these two (2) engineers will be finalized after Contract Award.

The training at site will consist of class room and on the job training. The class room training will be held for a period of at least four (4) months cumulative.

Owner will provide rooms for on-site training, furniture and all visual aids except video machines or equivalent which shall be provided by Bidder.

Bidder shall provide well organized training manuals for the trainees.

Training manuals and all the presentation of training program shall be in English.

(Last page of Subsection 2.13)

SUBSECTION - 2.14 SEISMIC REQUIREMENTS

2.14.1 SCOPE

The design of all equipment and systems shall be such that there is no loss of function and that safe conditions will prevail during and after a seismic disturbance based on the following requirement. “No loss of function” means that the integrity of the system is maintained such that after the seismic disturbance the equipment and system can be returned to service without repair or modification. The seismic forces considered in the design shall not be less than the minimum seismic forces for structures and equipment according to Tata Cara Perencanaan Ketahanan Gempa Untuk Bangunan Gedung SNI 03-1726-2012 .

2.14.2 CODES AND STANDARDS

The Bidder shall comply with the requirements of **Part 4, Subsection 2.5 "Codes and Standards"**.

The Codes and Standards of the following organizations extracted from the **Part 4, Subsection 2.5 “Codes and Standards”** are specifically applicable to the design, manufacture and testing of the work included in this Specification:

Tata Cara Perencanaan Ketahanan Gempa Untuk Bangunan Gedung SNI 03-1726-2012

2.14.3 BUILDINGS AND STRUCTURES HIGHER THAN 40 m

For buildings and structures whose height exceeds 40 m or of irregular shape the seismic design forces for all levels in the structure shall be determined from a dynamic analysis based on established principles of structural dynamics, which is a mandatory requirement of the latest Indonesia's Regulations and Codes as published by Ministry of Public Works, title “Tata Cara Perencanaan Ketahanan Gempa Untuk Bangunan Gedung” SNI 03-1726-2012 (Earthquake Design Regulation for Building) and "Pedoman Pembebanan untuk Gedung" (Indonesia's Loading Code). When the analysis is performed in two (2) orthogonal directions the effects due to an oblique earthquake shall also be

considered by combining the effect of the orthogonal directions using the square root of the sum of the square method.

The resulting base shear shall not be less than which would be obtained using the static analysis approach.

(Last page of Subsection 2.14)

SUBSECTION - 2.15 UNITS OF MEASUREMENTS AND SYMBOLS

2.15.1 UNITS OF MEASUREMENTS AND SYMBOLS

The following units of measurements and symbols shall be used for the entire project.

Quantity	Unit Description	Unit Symbols	Remark
Acceleration	Meter per square second	m/s ²	g: acceleration gravity
Area	Square meter Square centimeter Square millimeter	m ² cm ² mm ²	for cable cross section
Bolts & Nuts	Millimeter	mm	inch is not acceptable
Capacitance	Farad Microfarad	F μF	
Capacity	Mega Volt Ampere Kilo Volt Ampere Volt Ampere	MVA kVA VA	
Computer Storage Capacity	Byte Kilo Byte Mega Byte	byte kb Mb	
Concentration	Part per million Part per billion	ppm ppb	
Conductivity	Micro Siemens	μS	
Density (mass)	Kilogram per cubic meter Ton per cubic meter	kg/m ³ t/m ³	1 ton = 1000 kg
Electrical Current	Kilo Ampere Ampere Milli Ampere	kA A mA	
Electrical Energy	Kilowatt hours	kWh	
Fineness	Micro meter	μm	
Flow	Kilogram per hour	kg/h	e.g for steam/solid substance

Quantity	Unit Description	Unit Symbols	Remark
	Cubic meter per hour	m ³ /h	e.g for liquid or gas Normal = (N) for Natural gas at 1.013bar and 15°C
	Cubic meter per second	m ³ /s	
	Centimeter cubic per hour	cm ³ /h	
	Liter per second	l/s	
Mass (weight)	Ton (metric)	t	
	Kilogram	kg	
	Gram	g	
Frequency	Hertz	Hz	
GD ² Effect	Kilogram meter-square	kg m ²	
Heat	Kilo Joule	kJ	
Inductance	Henry	H	
Length/ thickness	Kilometre	km	No gauge, such as BWG for thickness
	Meter	m	
	Centimeter	cm	
	Millimeter	mm	
	Micrometre	µm	
Force	Newton	N	
	Kilo Newton	kN	
Moment (torque)	Ton meter	tm	
	kilogram meter	kgm	
Noise	Decibel	dB	
Pipe Rating	Schedule	Sch.	
Power	Watt	W	
	Kilowatt	kW	
	Megawatt	MW	
Pressure	Bar	bar	Absolute = (abs) or (not mentioned) gauge = (g)
Strength, stress	Kilo Newton per square meter	kN/m ²	
Pressure (Head)	Meter of water column	mWC	bar or milibar could also be accepted
	Millimeter of water column	mmWC	
	Millibar	mbar	
Printer Speed	Character per second		
	Line per minutes		

Quantity	Unit Description	Unit Symbols	Remark
Resistivity	Ohm	ohm, Ω	
Rotational Speed	Revolution per minute	rpm	
Temperature	Degrees of Celsius	$^{\circ}\text{C}$	Kelvin (K)
Time	Hour Minute Second	h min s	
Valves & Fittings Rating	Bar	bar	ANSI Rating lbs (#) is also acceptable but bar is preferable
Velocity	Kilometer per hour Meter per second Centimeter per second	km/h m/s cm/s	
Viscosity	Poise (dynamic) Stokes (kinematic)	P St	
Voltage	Kilo Volt Volt	kV V	
Volume	Litre	l	
	Cubic meter	m^3	
	Cubic centimeter	cm^3	or cc

(Last Page of Subsection 2.15)

SUBSECTION - 2.16 ENVIROMENTAL CONSIDERATIONS

2.16.1 GENERAL

The purpose of this subsection is to emphasize the environmental consideration required for the tropical climate and because proximity of the plant to the sea, high ambient temperatures in certain areas of the power plant, mold and fungus growth and risk of attack by vermin, termites etc. which could affect the life and operational reliability of Contractor's equipment and structures.

These environmental considerations are applicable to all equipment and structures furnished under the Contract. However they are especially important for the design and selection of control, instrumentation and electrical equipment.

2.16.2 ENVIRONMENTAL LEGAL AND REGULATORY

2.16.2.1 LAW AND REGULATION

The principal laws and regulations are to protect the environmental impacts of the propose project activity.

The construction of Muara Tawar add-on CCPP should comply of appropriate regulations, as follows:

1. Decree of State Minister of Environment No. 21, Year 2008 regarding Standard Quality of Non-moving Emission (As guidelines for standard quality of emission)
2. Government Regulation No. 41, Year 1999 regarding Control of Air Pollution (As guidelines in controlling air pollution)
3. Decree of State Minister of Environment No. 51, Year 1995 regarding Standard Quality of Waste Water (As guidelines for standard quality of waste water)
4. Decree of State Minister of Environment No. 51, Year 2004 regarding Standard Quality of Seawater (As guidelines for standard quality of seawater)

5. Decree of State Minister of Environment No. 12, Year 2006 regarding The Procedures and Condition for Standard of Waste Water Disposal to the Sea. (As guidelines for wastewater disposal to the sea)
6. Decree of State Minister of Environment No. 48, Year 1996 regarding Standard Quality of Noise (As guidelines for standard quality of noise)
7. Decree of Minister of Health No. 1405 , Year 2003 regarding Health Standard for Office and Industry Work field (As operational guidelines in this project)
8. Government Regulation No.18 jo No.85, Year 1999 regarding Management of Toxic and Hazardous Wastes (The activity plan must consider the management Toxic and Hazardous Wastes)
9. Government Regulation No. 74, Year 1999 regarding Management of Toxic and Hazardous Wastes.
10. Regulation of Chairman of Bapedal No. 01, Year 1995 regarding Management of Toxic and Hazardous (As guidelines for Management of Toxic and Hazardous Wastes)
11. Regional Regulation regarding environmental protection on West Java Province (If available)

In case before construction, if any changing regulation, the contractor should consider to the new regulation.

2.16.2.2 LENDER'S ENVIRONMENTAL REQUIREMENT

The Contractor should consider the Lender's Environmental Requirement, which is include on this project agreement.

2.16.3 ENVIRONMENTAL REQUIRMENT

2.16.3.1 ENVIRONMENTAL IMPACT ASSESSMENT REQUIREMENT

2.16.3.1.1 Muara Tawar add-on CCPP activity will impacts to the environment. According to The Environmental Impact Assessment (AMDAL) of Muara Tawar Power Plant, the project should complies to the instruction on Environmental Management Plan and Environmental Monitoring Plan on AMDAL document.

2.16.3.2 SOIL

Construction Period. Potential impacts to soil during the construction period may include ;

- (i) erosion of recently dredged soils,
- (ii) contamination from spillage of hydrocarbons and other hazardous wastes, and
- (iii) disposal of construction spoil.

The surface of the site may be particularly prone to water and wind erosion as it consists largely of un compacted, recently dredged soils

2.16.3.2.1 The Contractor should :

- (i) laying compacted topsoil as quickly as possible after construction at each area is completed, and
- (ii) allowing sediment to settle in sedimentation ponds and storm water channels prior to discharge into the Sea.

2.16.3.2.2 Soil contamination should be prevented by installing appropriate storage facilities for hazardous wastes and hydrocarbons on site.

2.16.3.2.3 All septic waste should be collected and removed from the site.

2.16.3.2.4 All hazardous and other waste will be removed from the site to approved waste disposal sites. There should be no on-site landfill.

2.16.3.2.5 Spoil generated from construction activities should be stockpiled, dewatered, and used for landscaping where possible or removed from the site to approved disposal areas.

2.16.3.2.6 Sediment fences should be installed around temporary spoil stockpiles to control erosion runoff.

2.16.3.3 WATER

Construction Phase

Potential impacts to water resources during the construction phase include;

- (i) disposal of septic wastewater generated by construction workers,
- (ii) disposal of equipment wash-down water/ acid cleaning and
- (iii) impacts on water reserves from over extraction.

2.16.3.3.1 All wastewater should be released off site to the wastewater treatment plant and treated, especially disposal of equipment wash down water or acid cleaning.

2.16.3.3.2 Septic wastewater generated during construction should be collected, treated in on-site facilities, and either disposed of in the nearby wastewater treatment facilities.

Operational Phase

The main potential impact to water resources during operation is the disposal of operational wastewater.

2.16.3.3.3 The wastewater should be released off site to the Wastewater Treatment Plant and treated to the level required by the Indonesia Standard of Wastewater Discharge Regulation of The State Minister of Environmental Decree No: **08, Year 2009 and No.: 05, year 2014**

If there is no statement in the table below, bidder shall comply with that regulation

No.	PARAMETER	UNITS	STANDARD
PHYSICS			
1.	Temperature	°C	40
2.	Dissolved solids	mg/l	2000
3.	Suspended solids	mg/l	100
CHEMICAL			
1.	PH		6-9
2.	Dissolved Iron (Fe)	mg/l	3
3.	Manganese (Mn)	mg/l	5
4.	Barium (Ba)	mg/l	3

ADD1
-IIA
-017

No.	PARAMETER	UNITS	STANDARD
5.	Copper (Cu)	mg/l	1
6.	Zinc (Zn)	mg/l	1
7.	Chrome hexavalent (Cr ⁶⁺)	mg/l	0.5
8.	Chrome total (Cr)	mg/l	0.5
9.	Cadmium (Cd)	mg/l	0.1
10.	Mercury (Hg)	mg/l	0.005
11.	Lead (Pb)	mg/l	1
12.	Tin (Sn)	mg/l	3
13.	Arsenic (As)	mg/l	0.5
14.	Selenium (Se)	mg/l	0.5
15.	Nickel (Ni)	mg/l	0.5
16.	Cobalt (Co)	mg/l	0.6
17.	Cyanide (CN)	mg/l	0.5
18.	Sulfide (H ₂ S)	mg/l	0.1
19.	Fluoride (F)	mg/l	3
20.	Free Chlorine (Cl ₂)	mg/l	0.5
21.	Free ammonia (NH ₃ -N)	mg/l	5
22.	Nitrate (NO ₃ -N)	mg/l	30
23.	Nitrite (NO ₂ -N)	mg/l	3
24.	BOD ₅	mg/l	150
25.	COD	mg/l	300
26.	Blue methyl	mg/l	10
27.	Phenol	mg/l	1
28.	Mineral oil and fat	mg/l	10
29.	TOC	mg/l	110
30	Phosphate (PO₄-)	mg/l	10

ADD1
-IIA
017

Notes:

- The pollutant contents in the waste water that comply to the above standard should not be adjusted by means of dilution with water obtained directly from the water source.
- The pollutants shall not degrade the chemical, physical and biological integrity of the water source.

2.16.3.3.4 The temperature outlet of the cooling system before released to the sea from the

outlet canal should not more than 40°C

The temperature outlet canal also comply to the Decree of State Minister of Environment No. 51, Year 2004 regarding Standard Quality of Seawater (As guidelines for standard quality of seawater)

The waste water temperature outlet canal model should be made by Contractor and expose to the Owner's.

2.16.3.4 AIR QUALITY

Construction Phase

The main impact to air quality during construction will be from increased dust levels from vehicle exhaust, construction machinery, foundation excavation, cement mixing, building construction, soil backfilling and minor access road construction.

- 2.16.3.4.1 There are any residential areas and Power generation in the vicinity of the Project. Mitigation should be made by paving all major access roads and using water trucks to suppress dust on unpaved roads.

2.16.3.4.2 NOISE

Construction Phase

During construction, noise will be generated from vehicular movement, piling, sand and aggregate processing, concrete mixing, excavation machinery, and construction. As the residential area nearest to the site is about ± 1 km away, moderate impact on residential areas should anticipated.

- 2.16.3.5.1 Equipment and machinery used for construction shall strictly conform to the Indonesia Noise Standards (Decree MLH No.48, Year 1996). All equipment will be properly maintained to minimize noise. Noise emissions from the existing site should be considered.
- 2.16.3.5.2 The workers should include standard occupational health and safety practices to relevant Indonesia standards.

Operational Phase

There are many noise sources during project operation, including gas turbines, steam turbines and cooling system, pumps, and air compressors, therefore the noise sources will be located mainly in buildings.

- 2.16.3.5.3 Noise mitigation measures will include enclosing the relevant equipment, noise suppression controls on relevant machinery, and seating particular facilities away from property boundaries.

Standard of Noise Level

The contractor should consider requirement of the PLN Standard No: SPLN 46 – 1, and SPLN 46-2 dated 1981. **and Keputusan Menteri Lingkungan Hidup No: KEP-48/MENLH-11-1996 tentang Baku Tingkat Kebisingan (Noise).**

ADD1
-IIA
-013

- a. Noise limit in the usage area.
- b. The Plant location is classified as industrial area.
- c. Contractor shall guarantee that the operating equipment will not produce a sound pressure level greater than 85 dB (A) when measured one meter from any point on any items of equipment or enclosure, including valves.

This requirement applies to the running equipment taking into account the noise effect from all other running equipment within the Plant.

- 2.16.3.5.4 Ongoing noise monitoring should be undertaken to ensure compliance. The noise monitoring equipment will be placed on site fence near the community house.

- 2.16.3.5.5 Noise modeling should be carried out in accordance to ensure compliance. The noise modeling could be indicates that the maximum noise level does not more than the requirement. Noise modeling results should be expose to the owner before.

2.16.3.5 SOLID WASTE AND HAZARDOUS MATERIALS

Construction Phase

- 2.16.3.6.1 Solid waste generated during construction should be collected, stored on site in appropriate storage facilities, and transported off site to approved disposal facilities.
- 2.16.3.6.2 No on-site landfills will be developed. Hazardous waste generated during construction should be collected and stored separately on site in approved facilities. Hazardous waste should be removed from the site to approved hazardous waste disposal facilities by a licensed waste transportation company.

2.16.3.6 SOCIOECONOMIC ENVIRONMENT

Construction Period. During project construction, the peak number of workers may reach 1,000 people. The influx of workers during construction may result in a friction with nearby community if the local people not involve in this project.

- 2.16.3.7.1 Where possible, the contractor should employee local people as a labor during construction as much as it possible.

(Last Page of Subsection 2.16)

SUBSECTION - 2.17 FUEL ANALYSIS

4.17.1 GENERAL

ADD1-IIA-014

Added Gas Composition and Fuel Oil as attached.

(Last Page of Subsection 2.17)

